

1. Diabetes Mellitus

Overview: Diabetes Mellitus is a chronic metabolic disorder characterized by elevated blood glucose levels due to defective insulin secretion, action, or both. It can lead to long-term complications if untreated.

Symptoms: Increased thirst, frequent urination, unexplained weight loss, fatigue, blurred vision, slow-healing wounds.

Causes: Genetic predisposition, obesity, sedentary lifestyle, and pancreatic beta-cell dysfunction.

Treatment: Lifestyle modification, oral hypoglycemic drugs, insulin therapy, regular monitoring of blood glucose.

Prevention: Maintain healthy weight, balanced diet, regular physical activity, and routine blood sugar screening.

2. Hypertension

Overview: Hypertension is a chronic condition of elevated arterial blood pressure that increases cardiovascular disease risk.

Symptoms: Often asymptomatic; headaches, dizziness, nosebleeds may occur in severe cases.

Causes: Genetics, obesity, high salt intake, stress, sedentary lifestyle.

Treatment: Antihypertensive medications, dietary adjustments, regular exercise, stress management.

Prevention: Limit salt, maintain healthy weight, exercise regularly, avoid alcohol and smoking.

3. Coronary Artery Disease

Overview: Coronary artery disease involves the narrowing of coronary arteries, reducing blood flow to the heart, leading to angina or heart attacks.

Symptoms: Chest pain, shortness of breath, fatigue, palpitations.

Causes: Atherosclerosis, high cholesterol, smoking, hypertension, diabetes, sedentary lifestyle.

Treatment: Lifestyle modification, medications (statins, antiplatelets), angioplasty, or coronary bypass surgery.

Prevention: Healthy diet, regular exercise, control of blood pressure and cholesterol, avoid smoking.

4. Asthma

Overview: Asthma is a chronic inflammatory airway disease causing intermittent airflow obstruction.

Symptoms: Wheezing, coughing, shortness of breath, chest tightness, especially at night or after exercise.

Causes: Genetic predisposition, allergies, respiratory infections, environmental pollutants.

Treatment: Inhaled corticosteroids, bronchodilators, leukotriene inhibitors, avoidance of triggers.

Prevention: Avoid allergens, smoking cessation, vaccination against respiratory infections.

5. Chronic Obstructive Pulmonary Disease (COPD)

Overview: COPD is a progressive lung disease characterized by airflow limitation and chronic respiratory symptoms.

Symptoms: Chronic cough, sputum production, dyspnea, frequent respiratory infections.

Causes: Long-term exposure to cigarette smoke, environmental pollutants, occupational dust, genetic factors.

Treatment: Smoking cessation, bronchodilators, corticosteroids, pulmonary rehabilitation, oxygen therapy in advanced cases.

Prevention: Avoid smoking and environmental pollutants, vaccinations for flu and pneumonia.

6. Tuberculosis

Overview: Tuberculosis is an infectious disease caused by *Mycobacterium tuberculosis*, primarily affecting the lungs.

Symptoms: Persistent cough, weight loss, fever, night sweats, hemoptysis.

Causes: Infection via airborne droplets from an active TB patient.

Treatment: Combination antibiotic therapy for 6–12 months (isoniazid, rifampicin, pyrazinamide, ethambutol).

Prevention: BCG vaccination, early detection, isolation of active cases, good ventilation.

7. Hepatitis B

Overview: Hepatitis B is a viral infection causing liver inflammation, which may progress to chronic liver disease.

Symptoms: Fatigue, nausea, jaundice, dark urine, abdominal pain.

Causes: Infection through blood, sexual contact, contaminated needles, or vertical transmission.

Treatment: Antiviral medications (tenofovir, entecavir), supportive care, regular liver monitoring.

Prevention: Vaccination, safe injection practices, avoiding unprotected sex, blood screening.

8. Hepatitis C

Overview: Hepatitis C is a viral liver infection that can become chronic, leading to cirrhosis and liver cancer.

Symptoms: Often asymptomatic initially; fatigue, jaundice, abdominal discomfort, dark urine.

Causes: Blood-borne transmission via contaminated needles, transfusions, or unsterile procedures.

Treatment: Direct-acting antivirals (DAAs) with high cure rates, liver monitoring.

Prevention: Avoid sharing needles, blood screening, safe medical practices.

9. HIV/AIDS

Overview: HIV is a virus that attacks the immune system, causing progressive immunodeficiency and AIDS if untreated.

Symptoms: Acute infection may present with flu-like symptoms; chronic infection leads to opportunistic infections and weight loss.

Causes: Sexual transmission, contaminated blood, mother-to-child transmission.

Treatment: Antiretroviral therapy (ART) to suppress viral load, treat opportunistic infections.

Prevention: Safe sex, condom use, ART for high-risk exposures, blood screening.

10. COVID-19

Overview: COVID-19 is a viral respiratory illness caused by SARS-CoV-2, ranging from mild infection to severe pneumonia.

Symptoms: Fever, cough, fatigue, loss of taste/smell, shortness of breath.

Causes: Infection via respiratory droplets, close contact, contaminated surfaces.

Treatment: Supportive care, oxygen therapy, antiviral drugs in severe cases, corticosteroids.

Prevention: Vaccination, masks, social distancing, hand hygiene.